

Socially Intelligent System: From Social Computing to Social Intelligence

Mingyu Kang^{1*}

¹ School of Cyber Science and Technology, University of Science and Technology of China, Hefei
230026, China

As argued by Jean-Jacques Rousseau, humans form societies because they believe that cooperation can generate greater productivity than individuals acting alone. But more important is how to build a strong and resilient society with connectivity, cooperation and social order. Thus, this paper proposes a concept of socially intelligent system here to demonstrate the necessity and feasibility of social computing and facilitate the emergence of social intelligence.

1) Social Computing:

In 1971, Project Cybersyn was developed during the presidency of Salvador Allende in Chile, with the purpose of constructing a distributed decision-making system to aid in the management of the national economy. Since the advent of computers, this was the first nation-level experiment to integrate computer, information and cybernetic technology into national management. The designer of this project, Stafford Beer, and his team tried to build a cybernetic system to achieve real-time synchronization between factories and government decision-makers. Although this project finally failed due to political circumstances, it inspired subsequent governments to explore computer, information and cybernetic technology to improve national governance. Decades later, it is called as social computing [1].

However, different from Project Cybersyn, today's advanced internet technology has enabled society to become interconnected as an integrated whole. Intelligent machines further transform the way human collaborate in production. We are gradually stepping into a human-machine society [2]. Thus, individual complexity only has little influence on the complexity of the massive social system, instead the network connections become more crucial. E.g., the two major discoveries made in the late 20-th century, Watts and Strogatz [3] found the small-world property, and then Barabási and Albert [4] found the scale-free property in social networks. Further, Wang and Chen [5] proposed a pinning strategy to control the whole scale-free network only with a few pinning nodes. These provide theoretical guarantees for conducting social experiments at a much larger scale.

Moreover, from the perspective of computing, the connected human-machine society can be viewed as a larger and more powerful computer. Although there is no evidence that humans can perform computations beyond computers, or any machine which is Turing completeness. But the computational efficiency of human-machine collaboration clearly exceeds that of a single human or a single machine. Thus, this comes to mind naturally, if we connect all humans and machines, that social computer could perhaps address more complex problems. And if we can clone and reproduce a human as a digital one, and even create digital humans, human society will evolve into a new form.

2) Computational Social Science:

Computational social science and social computing are usually confused with each other in Chinese context. However, computational social science is actually a branch of social science to investigate the interactive laws of

* kangmingyu@ustc.edu.cn

collective behaviors through informatics [6]. Before this seminal paper, there have been many social models, e.g., evolutionary game models [7], vicsek-like models [8] and complex networks [3,4]. Their common point is to explain how collective behaviors emerge through interactions, transitioning from disorder to order. With the internet and social medias, the interactions become more frequent and diverse in form. Many social data are generated, and this motivates us to translate the data into computable models and then use computer to autonomously analyze them. Thus, the current challenges lie in high-order interactions and dimension deduction for social modeling [9, 10], on the one hand.

On the other hand, the computational social science provides a new view of social systems by leveraging big data [11], and provides a pathway towards socially intelligent systems. With the artificial neural networks, most of social system functions can be represented and learned in theory, if data is sufficient. Especially, human language and semantics are also learned and simulated by large language models (LLMs), a large next-token predictor. This means an artificial intelligence (AI) can understand human society and think like a human from these data. Then, the characteristics of computational social science as a scientific discipline are gradually emerging. Recently, the vision of automating scientific research for social science through AI is taking shape [12]. This forms a closed loop, i.e., achieving social data, discovering laws from the data, applying them on human society and achieving new data for further discovery. However, due to the fact that AI is a “black box”, we need stronger methodologies to explain the causality in data for safeguard [13, 14].

3) *Cyber-Physical-Social System:*

With the networking and wireless communications, cyber-physical-social system (CPSS) provides another view to model the modern human society into coupling of cyber, physical and social layers [15]. In fact, there have been many research works on cyber-physical systems, which focus on industrial informatization. However, CPSS actually places greater emphasis on the human factors in system operation and management, as CPSS only incorporates an extra social layer compared with traditional cyber-physical system.

Before, without internet, human behaviors only influence surroundings. But the internet enlarges it beyond physical distance, and form a cyberspace to support a society [16]. In the cyber society, humans are connected by followee-follower networks and content recommender systems on social platforms, e.g., Weibo, Zhihu, Twitter, etc. Although the content generations are different on various platforms, the user behaviors are all strongly driven by attention. User attention is constrained by followee-follower networks and operated by content recommender systems, and then conversely influence physical world, even across vast spatial distance.

Moreover, there are some points that differ from physical society. First, human is without body, but only has multi-modal semantic contents and emotions. A human cannot touch and hurt another human’s body, and only can generate content that perhaps influences the other’s emotions. Second, a person may have multiple online identities, and even an identity cannot be traced to a person, e.g., it is a dog or a social bot. Thus, identities are hard to align, especially across different platforms, and human is not the unique content source. Third, the social relationships switch frequently, because the cost of network connections is low, and it is easily influenced by recommender systems. In contrast, the social relationships in physical society change slowly and are largely determined by spatial location.

4) *Social Simulation:*

Do digital humans dream of digital sheep? In cyberspace, there are only multi-modal data carrying both semantic contents and emotions. We cannot distinguish two people through physical entity. Thus, we need to know how much data and what rules are required to define a human, or define a society.

In 1987, Laird et al. [17] introduced SOAR project, a symbolic cognitive framework for AI agent. Then, Minar et al. [18] introduced SWARM project to build multi-agent systems for simulating collective behaviors. Around the

same time, the science of complex networks [3, 4] also began to emerge. In 2023, with the emergence of chatbots, i.e. ChatGPT, Park et al. [19] introduced Stanford Town AI project, that uses chatbots equipped with LLMs to simulate human-like behaviors in open environment.

However, such social simulations face a fundamental problem, i.e., how to quantify and measure the difference between the real world and simulations. Thus, we here provide a causal criterion. For example, given a social network $G(V, E)$, where $V = \{X_1, \dots, X_5, T, Y\}$, and the network is shown in Fig. 1. If we treat $T = t$ at time $s - 2$, like a social event, the estimation of target Y at time s can be defined as

$$\hat{y}_s = \mathbb{E}[Y_s | \text{do}(T_{s-2} = t)] = \int P(Y_s = y | \text{do}(T_{s-2} = t)) dy, \quad (1)$$

where $P(\cdot)$ is probability distribution, and $\text{do}(\cdot)$ is do-calculus operator [13, 20]. Then, we can transform the social network into a spatio-temporal Bayesian network and adjust the treated distribution as

$$\begin{aligned} & P(Y_s = y | \text{do}(T_{s-2} = t)) \\ &= \sum_{x_2, x_5} P(Y_s = y | X_{2,s-1} = x_2, X_{5,s-1} = x_5) \\ & \quad \times P(X_{2,s-1} = x_2 | T_{s-2} = t) P(X_{5,s-1} = x_5 | T_{s-2} = t). \end{aligned} \quad (2)$$

Then, the difference can be measured by comparing the unbiased $\hat{y}_s$ with the ground truth

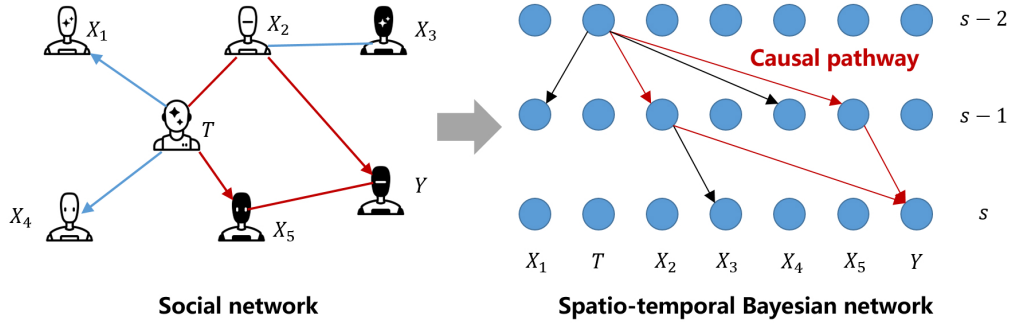


Figure 1. Diagram of causal criterion for social simulation.

Thus, reproducing a social event requires a specific algorithm to accurately simulate the event in a social simulation platform. If the platform can simulate all social events, it is called universal. Of course, there also some studies demonstrating that the social bots (a simulation algorithm) have obvious difference from real human from the perspective of entropy [21]. And the LLM-driven social bots cannot meet the human intelligence [14]. All these point to a question, can data and rules build a completely digital society.

5) Embodied Social Intelligence:

If data and rules cannot build a completely digital society, that means intelligence cannot exist independently of physical entities, we need to embody an AI with a physical body. Thus, it is natural to think of embodied intelligence, that emphasizes interactions with environments [22, 23].

In classical controls, an intelligent machine with predefined rules try to reach a table state, and it hardly produce intelligence by generalizing rules to new tasks. Thus, some works try to use reinforcement learning, an optimal, model predictive, adaptive control, to enable an AI to interact and learn from open environment in a data-driven manner. The key to data-driven control lies in data acquirement and task generalization through interactions. However, in some industrial scenarios, e.g., power systems or communication systems, they do not allow repeated trial and error, but this is essential for most of the current data-driven manners. To address this issue, one way is to embed expert knowledge in prior, but knowledge is limited. More importantly, AI that cannot operate independently

of humans appear to be limited. Another way is to enable AI to learn from its own experience rather than data human made [24]. This leads to the embodied social intelligence, that is, an embodied AI as an individual can join society.

On the other hand, from the perspective of CPSS, there are three layers, social layer (data & rules) $\leftrightarrow$ cyber layer (communications) $\leftrightarrow$ physical layer (entities). If without cyber and physical layer, i.e., a completely digital society, the social layer has nothing more than a heap of data. The rules partition the data into individuals and define the interactions among individuals. But with physical entities, they are naturally partitioned into individuals under the constraints of communication rules by the cyber layer. Thus, if we continuously improve the communication efficiency, the real-world human society will increasingly interconnected as a whole. This will facilitate more frequent and complex interactions and giving rise to a higher-level social intelligence.

6) *Symbolic Grounding of Social Intelligence:*

If social intelligence is grounded in physical entities, it requires symbolic grounding, otherwise it is perhaps symbolic manipulation only without true semantic understanding. In 1950, Turing [25] proposed a imitation game to test machine intelligence. However, John Searle then provided a Chinese room example to show the challenges, that is, an AI can talk with a human only with symbolic manipulation and that human cannot tell whether it is a machine. Recently, LLM has been regarded as a modern case of Chinese room [26].

Thus, how does social intelligence achieve symbolic grounding? Taddeo and Floridi [27] critically examined eight views of symbolic grounding and argued that they all violate the zero semantical commitment condition, that is, the interpretation of symbols must be intrinsic to the symbol system itself, rather than extrinsic. The key issue lies in the fact that the symbols used by AI, whatever the tasks, intents, and meanings, they are all aligned with humans, or traced back to humans. They do not require self-preservation and survival in the physical world as a foundational rule for grounding.

However, it is reasonable for social intelligence to meet the zero semantical commitment condition, from the view of interactions and evolution. The reason is that, an AI with symbolic grounding must have a foundational rule to ensure self-preservation and its own survival, otherwise it would not persist in the physical world. Thus, the survival principle is a necessary condition for the existence of social intelligence, preceding symbolic grounding.

Acknowledgment: This work was supported by the China Postdoctoral Science Foundation (Grant No. 2026M791870).

References

- [1] F.-Y. Wang, K. M. Carley, D. Zeng, and W. Mao, “Social computing: From social informatics to social intelligence,” *IEEE Intelligent Systems*, vol. 22, no. 2, pp. 79–83, 2007.
- [2] M. Tsvetkova, T. Yasseri, N. Pescetelli, and T. Werner, “A new sociology of humans and machines,” *Nature Human Behaviour*, vol. 8, no. 10, pp. 1864–1876, 2024.
- [3] D. J. Watts and S. H. Strogatz, “Collective dynamics of ‘small-world’ networks,” *Nature*, vol. 393, no. 6684, pp. 440–442, 1998.
- [4] A.-L. Barabási and R. Albert, “Emergence of scaling in random networks,” *Science*, vol. 286, no. 5439, pp. 509–512, 1999.
- [5] X. F. Wang and G. Chen, “Pinning control of scale-free dynamical networks,” *Physica A: Statistical Mechanics and its Applications*, vol. 310, no. 3-4, pp. 521–531, 2002.
- [6] D. Lazer, A. Pentland, L. Adamic, S. Aral, A.-L. Barabási, D. Brewer, N. Christakis, N. Contractor, J. Fowler, M. Gutmann, T. Jebara, G. King, M. Macy, D. Roy, and M. V. Alstyne, “Computational social science,” *Science*, vol. 323, no. 5915, pp. 721–723, 2009.
- [7] J. Maynard Smith and G. R. Price, “The logic of animal conflict,” *Nature*, vol. 246, pp. 15–18, 1973.
- [8] T. Vicsek, A. Czirók, E. Ben-Jacob, I. Cohen, and O. Shochet, “Novel type of phase transition in a system of self-driven particles,” *Physical Review Letters*, vol. 75, pp. 1226–1229, 1995.
- [9] Y. Gao, S. Ji, X. Han, and Q. Dai, “Hypergraph computation,” *Engineering*, vol. 40, pp. 188–201, 2024.
- [10] J. Gao, “Intrinsic simplicity of complex systems,” *Nature Physics*, vol. 20, pp. 184–185, 2024.
- [11] W. Yu, D. Chen, H. Liu, H. Wang, J. Cao, Z. Di, X. Duan, B. Jia, X. Ding, and Y. Hong, “Systems science in the new era: intelligent systems and big data,” *Science China Information Sciences*, vol. 67, no. 3, p. 136201, 2024.
- [12] G. Tie, J. Shi, D. Song, Y. Huang, Z. Sheng, X. Zhou, D. Liu, P. Zhou, Y. Chen, R. Xu, L. He, Q. Wen, M. Li, C. Lu, S. Li, P. Xie, Y. Yuan, R. Meng, L. Xing, L. Sun, C. Xiong, P. S. Yu, and J. Gao, “Autoresearch ai: Towards ai-powered research automation for scientific discovery,” 2026. [Online]. Available: <https://arxiv.org/abs/2605.23204>
- [13] M. Kang, D. Chen, Z. Pu, J. Gao, and W. Yu, “Spatio-temporal graphical counterfactuals: an overview,” *Science China Information Sciences*, vol. 69, no. 4, p. 141201, 2026.
- [14] Y. Xie, L. Liang, S. Li, Y. Lu, Z. Xiao, M. Shi, J. Huang, M. Wang, and Y. Xie, “Evaluating the statistical realism of llm-generated social science data,” *Proceedings of the National Academy of Sciences*, vol. 123, no. 19, p. e2538145123, 2026.

- [15] Y. Zhou, F. R. Yu, J. Chen, and Y. Kuo, “Cyber-physical-social systems: A state-of-the-art survey, challenges and opportunities,” *IEEE Communications Surveys & Tutorials*, vol. 22, no. 1, pp. 389–425, 2020.
- [16] L. Pan, C.-Y. Wang, F. Zhou, and L. Lü, “Complexity of social media in the era of generative ai,” *National Science Review*, vol. 12, no. 1, p. nwae323, 2025.
- [17] J. E. Laird, A. Newell, and P. S. Rosenbloom, “Soar: An architecture for general intelligence,” *Artificial Intelligence*, vol. 33, no. 1, pp. 1–64, 1987.
- [18] N. Minar, R. Burkhart, C. Langton, and M. Askenazi, “The swarm simulation system: A toolkit for building multi-agent simulations,” Santa Fe Institute, Santa Fe, NM, USA, Working Paper, 1996. [Online]. Available: <https://www.santafe.edu/research/results/working-papers/the-swarm-simulation-system-a-toolkit-for-building>
- [19] J. S. Park, J. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, “Generative agents: Interactive simulacra of human behavior,” in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.
- [20] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge, UK: Cambridge University Press, 2009.
- [21] S. Jiang, Y. Xie, X.-K. Xu, and L. Lü, “Entrobot: Entropy-based modeling of account behavior for social bot detection,” *IEEE Transactions on Computational Social Systems*, 2026. [Online]. Available: <https://ieeexplore.ieee.org/abstract/document/11575568>
- [22] Y. LeCun, “A path towards autonomous machine intelligence,” 2022, version 0.9.2, 2022-06-27. [Online]. Available: <https://openreview.net/pdf?id=BZ5a1r-kVsf>
- [23] D. Hendrycks, D. Song, C. Szegedy, H. Lee, Y. Gal, E. Brynjolfsson, S. Li, A. Zou, L. Levine, B. Han, J. Fu, Z. Liu, J. Shin, K. Lee, M. Mazeika, L. Phan, G. Ingebrechtsen, A. Khoja, C. Xie, O. Salaudeen, M. Hein, K. Zhao, A. Pan, D. Duvenaud, B. Li, S. Omohundro, G. Alfour, M. Tegmark, K. McGrew, G. Marcus, J. Tallinn, E. Schmidt, and Y. Bengio, “A definition of agi,” 2025. [Online]. Available: <https://arxiv.org/abs/2510.18212>
- [24] D. Silver and R. S. Sutton, “Welcome to the era of experience,” DeepMind Technical Note, 2025. [Online]. Available: <https://storage.googleapis.com/deepmind-media/Era-of-Experience%20The%20Era%20of%20Experience%20Paper.pdf>
- [25] A. M. Turing, *Computing Machinery and Intelligence (1950)*. Dordrecht: Springer Netherlands, 2009, pp. 23–65.
- [26] R. Gubelmann, “Pragmatic norms are all you need – why the symbol grounding problem does not apply to LLMs,” in *Empirical Methods in Natural Language Processing*, Y. Al-Onaizan, M. Bansal, and Y.-N. Chen, Eds. Association for Computational Linguistics, 2024.
- [27] M. Taddeo and L. Floridi, “Solving the symbol grounding problem: A critical review of fifteen years of research,” *Journal of Experimental & Theoretical Artificial Intelligence*, vol. 19, no. 4, pp. 419–445, 2007.